



INFINITY TUTORS

PHY 122

ELECTRICITY AND MAGNETISM, VIBRATIONS AND WAVES

TUTORIAL 3.

ELECTRIC CHARGE AND ELECTRIC FIELD

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Instructions

1. Answer all the questions
 2. Show your work clearly
 3. Verify your work with the Infinity Tutors
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SOME IMPORTANT CONSTANTS

$$1 \text{ mC} = 10^{-3} \text{C}, \quad 1 \mu \text{C} = 10^{-6} \text{C}, \quad 1 \text{ nC} = 10^{-9} \text{C}$$

$$k_e = 8.99 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \quad |e| = 1.6 \times 10^{-19} \text{C}$$

1 Charge & Electric Force

1. Define electric Charge [2]
2. What is the magnitude of the electric force of attraction between an iron nucleus ($q = +26e$) and its innermost electron if the distance between them is $1.5 \times 10^{-12} \text{ m}$? [5]
3. Calculate the magnitude and direction of the Coulomb force on $6\mu\text{C}$ charge shown in Figure 1 [7]

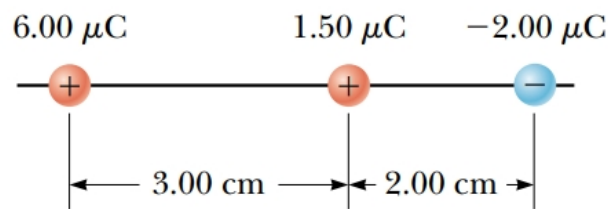


Figure 1:

4. State the Coulomb's law [2]

5. Three charges are arranged as shown in Figure 2 Find the magnitude and direction of the electrostatic force on the charge at the origin [8]

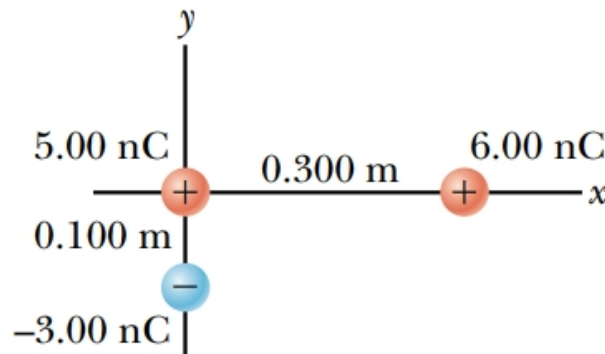


Figure 2:

6. Three point charges are located at the corners of an equilateral triangle as in Figure 3. Find the magnitude and direction of the net electric force on the 2.00 μC charge [9]

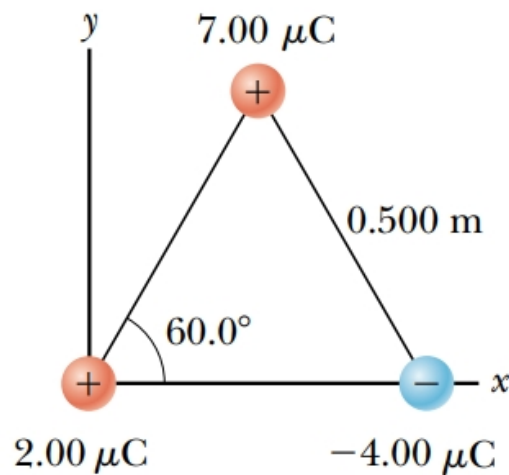


Figure 3:

7. In fission, a nucleus of Uranium-235, which contains 92 protons, divides into two smaller spheres, each having 46 protons and a radius of $5.9 \times 10^{-15} \text{ m}$. What is the magnitude of the repulsive electric force pushing the two spheres apart? [5]

8. Two point charges are located on the $x - axis$. They are both positive, but the one located at $x = 0$ has a charge q while the other located at $x = L$ has a charge of $4q$. If a third charge is placed on the $x - axis$ in between the two charges so that the net force on the third charge is zero, determine the location of the third charge. [9]
9. What is the force on the $2.00\mu C$ charge placed at the center of the square shown below? [9]

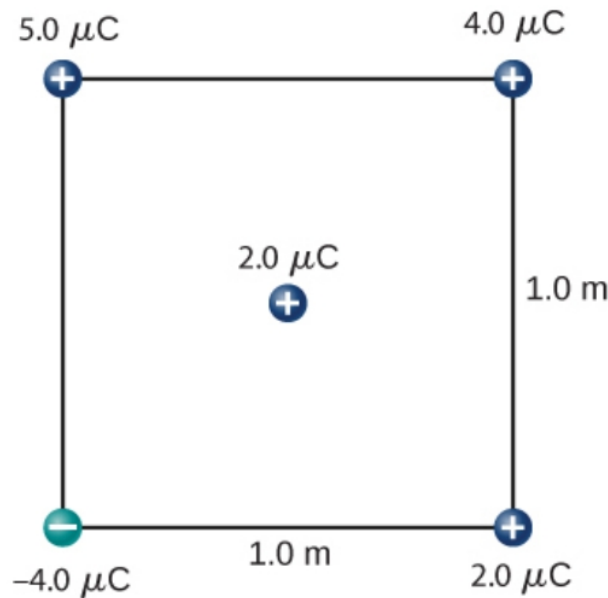


Figure 4:

10. Two small metallic spheres, each of mass $m = 0.20g$, are suspended as pendulums by light strings from a common point as shown in Figure 5. The spheres are given the same electric charge, and it is found that they come to equilibrium when each string is at an angle of $\theta = 5.0^\circ$ with the vertical. If each string has length $L = 30cm$, what is the magnitude of the charge on each sphere? [9]

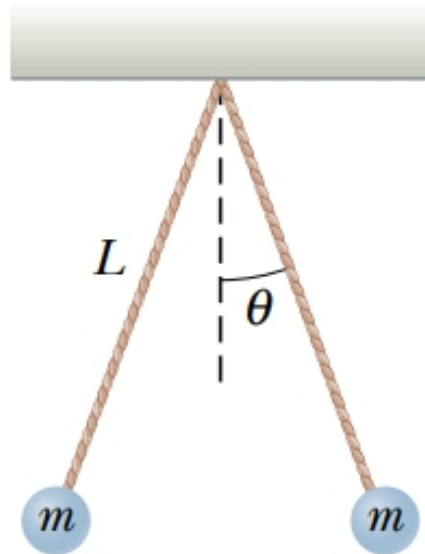


Figure 5:

2 Electric Field

1. Define electric field [2]
2. Three charges are at the corners of an equilateral triangle, as shown in Figure 6. Calculate the electric field at a point midway between the two charges on the x-axis. [8]

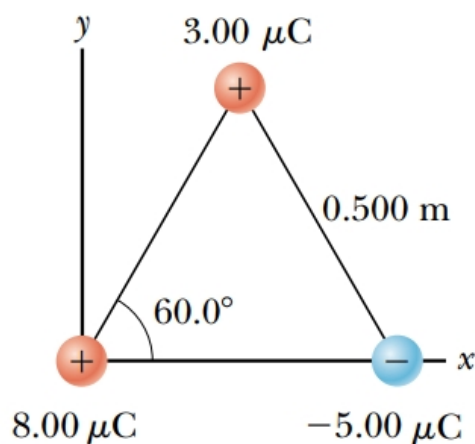


Figure 6:

3. Three point charges are aligned along the x-axis as shown in Figure 7 . Find the electric field at the position $x = +2\text{m}$, $y = 0$ [8]

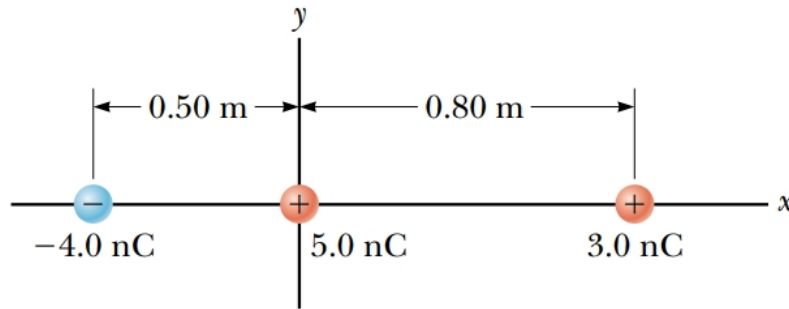


Figure 7:

4. The electric field everywhere on the surface of a charged sphere of radius 0.230 m a magnitude of 575 N/C and points radially outward from the center of the sphere. What is the net charge on the sphere? [5]
5. In Figure 8 determine the point (other than infinity) at which the total electric field is zero. [8]

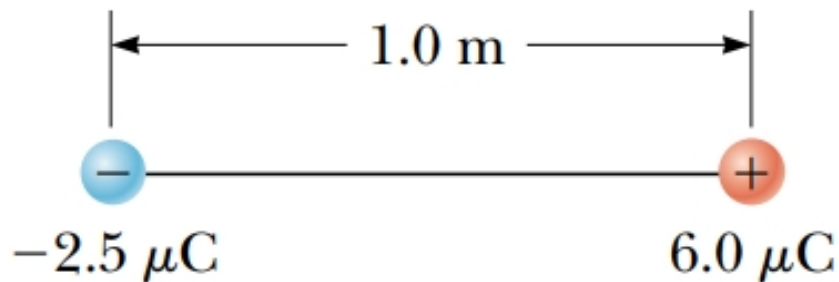


Figure 8:

” Attracting knowledge, one electric field at a time: Valentine’s Day!”
